

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

Practical - VIII

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8. Implement the following EJB applications with different types of Beans

a. Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

index.html :

```
<!DOCTYPE html>
<html>
<head>
  <title>Hit Count Demo</title>
</head>
<body>
<form action="ShowHitCountServlet" method="post">
  <input type="submit"
    value="Take Hit Count demo with EJB Session Singleton Bean">
</form>
</body>
</html>
```

STHitCntBean.java :

```
package EJPracts;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "ShowHitCountServlet",
  urlPatterns = {"/ShowHitCountServlet"})
public class ShowHitCountServlet extends HttpServlet {

  @EJB
  private STHitCntBean hitCountBean;

  protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    int count = hitCountBean.incrHitCount();
```

```

        out.println("<html>");
        out.println("<body>");
        out.println("<h2>Servlet Hit Count: " + count + "</h2>");
        out.println("</body>");
        out.println("</html>");
    }
}

```

ShowHitCountServlet.java :

```

package EJPracts;

import javax.ejb.Singleton;

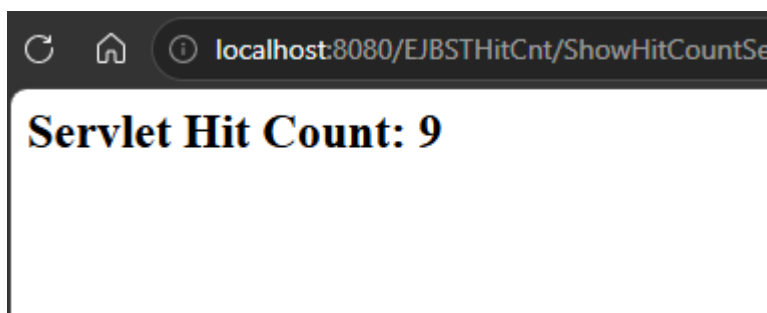
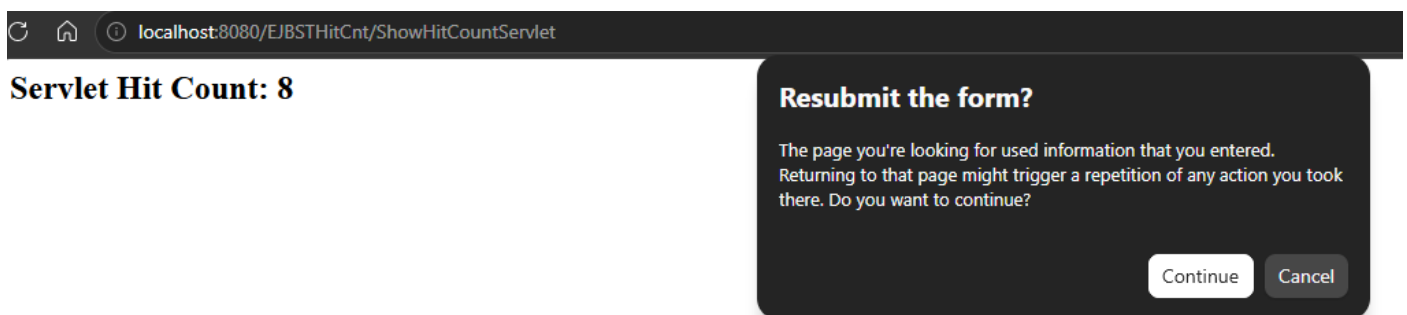
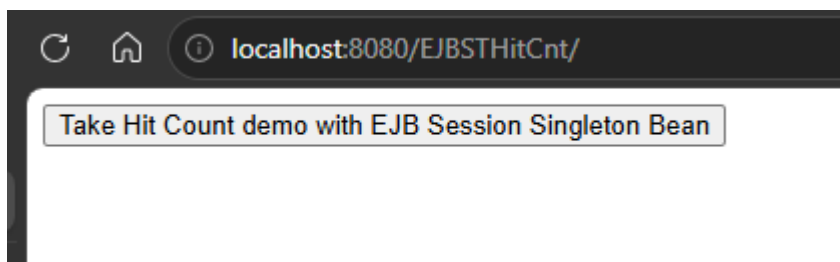
@Singleton
public class STHitCntBean {

    private int count;

    public synchronized int incrHitCount() {
        return count++;
    }
}

```

OUTPUT:-



b. Develop simple visitor Statistics application using Message Driven Bean [MDB].

Index.jsp

```

<%@page import="javax.jms.JMSEException"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.Session"%>
<%@page import="javax.jms.Connection"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.ConnectionFactory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Message Driven Bean</title>
</head>
<body style="background-color:gray;">
<h1>Welcome To Message Driven Bean</h1>
<%
    ConnectionFactory connFact = null;
    Queue queue = null;
    Connection conn = null;
    Session mySess = null;
    MessageProducer msgProducer = null;
    TextMessage msg = null;
    try {
        InitialContext ctx = new InitialContext();
        queue = (Queue) ctx.lookup("jms/Queue");
        connFact = (ConnectionFactory)
            ctx.lookup("jms/QueueFactory");
        conn = connFact.createConnection();
        mySess = conn.createSession(
            false,
            Session.AUTO_ACKNOWLEDGE
        );
        msgProducer = mySess.createProducer(queue);
        msg = mySess.createTextMessage();
        msg.setText(request.getRemoteAddr());
        msgProducer.send(msg);
        conn.close();
    } catch (JMSEException e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    } catch (Exception e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    }
%>
</body>

</html>

```

Sql:-

USE reg;
Database changed

```
CREATE TABLE visitorstat (
    visitdate TIMESTAMP NOT NULL,
    host VARCHAR(45),
    visitcount INT,
    PRIMARY KEY (host)
);
```

```
SELECT * FROM visitorstat;
```

MsgDrvnBean.java

```
package EJPract;
import javax.annotation.Resource;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.ejb.MessageDrivenContext;
import javax.jms.JMSException;
import javax.jms.Message;
import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(
    activationConfig = {
        @ActivationConfigProperty(
            propertyName = "destinationType",
            propertyValue = "javax.jms.Queue"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationLookup",
            propertyValue = "jms/Queue"
        )
    }
)
public class MsgDrvnBean implements MessageListener {
    @EJB
    VStatelessBean vs;
    @Resource
    private MessageDrivenContext mdc;

    public MsgDrvnBean() {
    }
    @Override
    public void onMessage(Message message) {
        try {
            if (message instanceof TextMessage) {
                TextMessage msg =
                    (TextMessage) message;
                vs.addVisitor(msg.getText());
            }
        }
        catch (JMSException e) {
            mdc.setRollbackOnly();
        }
    }
}
```

VStatelessBean.java

```
package EJPract;
import java.sql.Connection;
```

```

import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
    private Connection conn = null;
    private ResultSet rs;
    private Statement stmt = null;
    private String query = null;
    @PostConstruct
    public void connect() {
        try {
            Class.forName("com.mysql.jdbc.Driver");
            String connStr =
                "jdbc:mysql://localhost/reg";
            conn = DriverManager.getConnection(
                connStr,
                "root",
                "root"
            );

            System.out.println("Connection Success.");
        } catch (ClassNotFoundException | SQLException e) {
            System.err.println("Connection failed: " + e.getMessage());
        }
    }
    @PreDestroy
    public void disconnect() {
        try {
            conn.close();
            System.out.println(
                "Connection closed."
            );
        }
        catch (SQLException e) {

            System.err.println(
                "Not able to close connection: "
                + e.getMessage()
            );
        }
    }
    public void addVisitor(String host) {
        try {
            stmt = conn.createStatement();
            query =
                "INSERT INTO visitorstat(host,visitcount) " +
                "VALUES('" + host + "',1)";

            stmt.executeUpdate(query);
        }
        catch (SQLException e) {
            try {
                stmt = conn.createStatement();
            }

```

```

        query =
            "UPDATE visitorstat " +
            "SET visitcount = visitcount + 1 " +
            "WHERE host = '" + host + "'";
        stmt.executeUpdate(query);
    }
    catch (SQLException ex) {

        System.err.println(
            "Cannot update: "
            + ex.getMessage()
        );
    }
}
}
}
}

```

OUTPUT:-

Welcome To Message Driven Bean

```

mysql> USE reg;
Database changed
mysql> SELECT * FROM visitorstat;
+-----+-----+-----+
| visitdate          | host                | visitcount |
+-----+-----+-----+
| 2026-08-29 11:24:28 | 0:0:0:0:0:0:0:1    | 18         |
+-----+-----+-----+
1 row in set (0.00 sec)

```

c. Develop simple Marks Entry Application to demonstrate accessing Database using EJB.

CODE:-

Index.jsp

```

<%@page import="java.util.ArrayList"%>
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="EJPract.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!
    private static MarksEntryBean marksEntry;
    public void jspInit() {
        try {
            InitialContext ic = new InitialContext();
            marksEntry = (MarksEntryBean)
                ic.lookup(
                    "java:global/EJBMarksEntry/MarksEntryBean"
                );
        }
        catch (Exception ex) {
            System.out.println(
                "Database Error " + ex.getMessage()
            );
        }
    }
%>
<%
    if (request.getParameter("insertMarks") != null) {

```

```

String name, marks1, marks2, marks3;
name = request.getParameter("stuName");
marks1 = request.getParameter("sub1");
marks2 = request.getParameter("sub2");
marks3 = request.getParameter("sub3");
marksEntry.addMarks(
    name,
    marks1,
    marks2,
    marks3
);
}
%>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Marks Entry</title>
</head>
<body>
<h2 style="text-align:center;">
    Enter Details
</h2>
<br>
<form name="result" method="post">
<table>
<tr>
    <td>Enter Students Name</td>
    <td>
        <input type="text"
            name="stuName"
            value="">
    </td>
</tr>
<tr>
    <td>Enter Subject 1 Marks</td>
    <td>
        <input type="text"
            name="sub1"
            value="">
    </td>
</tr>
<tr>
    <td>Enter Subject 2 Marks</td>
    <td>
        <input type="text"
            name="sub2"
            value="">
    </td>
</tr>
<tr>
    <td>Enter Subject 3 Marks</td>
    <td>
        <input type="text"
            name="sub3"
            value="">
    </td>
</tr>

```

```

<tr>
  <td colspan="2">
    <input type="submit"
      value="insertMarks"
      name="insertMarks">
  </td>
</tr>
</table>
</form>
</body>
</html>

```

MarksEntryBean.java:-

```

package EJPract;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
import javax.ejb.Stateful;
@Stateful
public class MarksEntryBean {
    List<String> details;
    String stuName, sub1, sub2, sub3;
    private Connection conn = null;
    private ResultSet rs;
    private Statement stmt = null;
    private String query = null;
    public void createConnection() {
        try {
            Class.forName("org.gjt.mm.mysql.Driver");
            String connStr =
                "jdbc:mysql://localhost/reg";
            conn = DriverManager.getConnection(
                connStr,
                "root",
                "root"
            );
        } catch (ClassNotFoundException | SQLException e) {
            System.err.println(
                "Database Error " + e.getMessage()
            );
        }
    }
    public void addMarks(
        String stuname,
        String marks1,
        String marks2,
        String marks3) {
        createConnection();
        try {
            stmt = conn.createStatement();
            query =
                "INSERT INTO marks" +
                "(stuName,sub1,sub2,sub3)" +
                "VALUES(" +
                stuname + "," +
                marks1 + "," +
                marks2 + "," +
                marks3 + ")";

```

```
        stmt.executeUpdate(query);
    }
    catch (SQLException e) {
        System.err.println(
            "Failed to insert. "
            + e.getMessage()
        );
    }
}
```

OUTPUT:-

```
mysql> SELECT * FROM marks;
+----+-----+-----+-----+-----+
| id | stuName | sub1 | sub2 | sub3 |
+----+-----+-----+-----+-----+
| 1  | Subu    | 100  | 69   | 100  |
| 2  | Khushi  | 100  | 100  | 100  |
| 3  | Ishita  | 100  | 100  | 100  |
| 4  | Priti   | 1000 | 100  | 100  |
+----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```